Deep Learning for NLP

Generative Pre-Trained Transformers
GPT-3: Tasks & Performance



Learning goals

- Overview on relevant tasks and benchmarks
- GPT-3's performance in X-shot settings

LAMBADA TASK

Context →	Fill in blank: She held the torch in front of her. She caught her breath. "Chris? There's a step." "What?" "A step. Cut in the rock. About fifty feet ahead." She moved faster. They both moved faster. "In fact," she said, raising the torch higher, "there's more than a ____. ->
Target Completion →	step

▸ Source: Brown et al., 2020

PERFORMANCE ON LAMBADA

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0[a]	8.63[b]	**91.8**[c]	**85.6**[d]
GPT-3 Zero-Shot	**76.2**	**3.00**	83.2	78.9
GPT-3 One-Shot	**72.5**	**3.35**	84.7	78.1
GPT-3 Few-Shot	**86.4**	**1.92**	87.7	79.3

▸ Source: Brown et al., 2020

“CLOSED BOOK” QUESTION ANSWERING (QA)

Context →	Q: ‘Nude Descending A Staircase’ is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

▸ Source: Brown et al., 2020

PERFORMANCE ON THE CLOSED-BOOK QA

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP+20]	**44.5**	**45.5**	**68.0**
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	**68.0**
GPT-3 Few-Shot	29.9	41.5	**71.2**

▸ Source: Brown et al., 2020

PERFORMANCE ON MACHINE TRANSLATION

Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	**45.6**[a]	35.0 [b]	**41.2**[c]	40.2[d]	**38.5**[e]	**39.9**[e]
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ+19]	37.5	34.9	28.3	35.2	35.2	33.1
mBART [LGG+20]	-	-	29.8	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	39.2	29.7	40.6	21.0	39.5

▸ Source: Brown et al., 2020

WINOGRAD TASK

Correct Context →	Grace was happy to trade me her sweater for my jacket. She thinks the sweater
Incorrect Context →	Grace was happy to trade me her sweater for my jacket. She thinks the jacket
Target Completion →	looks dowdy on her.

▸ Source: Brown et al., 2020

PERFORMANCE ON WINOGRAD TASK

Setting	Winograd	Winogrande (XL)
Fine-tuned SOTA	**90.1**[a]	**84.6**[b]
GPT-3 Zero-Shot	88.3*	70.2
GPT-3 One-Shot	89.7*	73.2
GPT-3 Few-Shot	88.6*	77.7

▸ Source: Brown et al., 2020

ARC TASK

- Multipe-Choice questions from 3rd to 9th grad science exams
- *Challenge* version: Filtered to questions that simple statistical or information retrieval methods are unable to correctly answer

Context →	Question: George wants to warm his hands quickly by rubbing them. Which skin surface will produce the most heat? Answer:
Correct Answer →	dry palms
Incorrect Answer →	wet palms
Incorrect Answer →	palms covered with oil
Incorrect Answer →	palms covered with lotion

▸ Source: Brown et al., 2020

PERFORMANCE ON ARC TASK

Setting	PIQA	ARC (Easy)	ARC (Challenge)	OpenBookQA
Fine-tuned SOTA	79.4	**92.0**[KKS+20]	**78.5**[KKS+20]	**87.2**[KKS+20]
GPT-3 Zero-Shot	**80.5***	68.8	51.4	57.6
GPT-3 One-Shot	**80.5***	71.2	53.2	58.8
GPT-3 Few-Shot	**82.8***	70.1	51.5	65.4

RACE TASK

Context →	Article: Informal conversation is an important part of any business relationship.Before you start a discussion,however,make sure you understand which topics are suitable and which are considered taboo in a particular culture. Latin Americans enjoy sharing information about their local history, art and customs.You may expect questions about your family,and be sure to show pictures of your children.You may feel free to ask similar questions of your Latin American friends.The French think of conversation as an art form,and they enjoy the value of lively discussions as well as disagreements. For them,arguments can be interesting and they can cover pretty much or any topic ---- as long as they occur in are respectful and intelligent manner. In the United States,business people like to discuss a wide range of topics,including opinions about work,family,hobbies,and politics. In Japan,China,and Korea,however,people are much more private.They do not share much about their thoughts,feelings,or emotions because they feel that doing so might take away from the harmonious business relationship they're trying to build.Middle Easterners are also private about their personal lives and family matters.It is considered rude,for example,to ask a businessman from Saudi Arabia about his wife or children. As a general rule,it's best not to talk about politics or religion with your business friends.This can get you into trouble,even in the United States,where people hold different religious views.In addition,discussing one's salary is usually considered unsuitable.Sports is typically a friendly subject in most parts of the world,although be careful not to criticize national sport.Instead,be friendly and praise your host's team. Q: What shouldn't you do when talking about sports with colleagues from another country? A: Criticizing the sports of your colleagues' country. Q: Which is typically a friendly topic in most places according to the author? A: Sports. Q: Why are people from Asia more private in their conversation with others? A: They don't want to have their good relationship with others harmed by informal conversation. Q: The author considers politics and religion _ . A:
Correct Answer →	taboo
Incorrect Answer →	cheerful topics
Incorrect Answer →	rude topics
Incorrect Answer →	topics that can never be talked about

PERFORMANCE ON RACE TASK

Setting	CoQA	DROP	QuAC	SQuADv2	RACE-h	RACE-m
Fine-tuned SOTA	**90.7**[a]	**89.1**[b]	**74.4**[c]	**93.0**[d]	**90.0**[e]	**93.1**[e]
GPT-3 Zero-Shot	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3 One-Shot	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3 Few-Shot	85.0	36.5	44.3	69.8	46.8	58.1

PERFORMANCE ON SUPERGLUE TASK

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

SUPERGLUE

- BoolQ
- CB (true/false/neither)
- COPA
- RTE (similar to natural language inference)
- WiC
- WSC
- MultiRC (true/false)
- ReCoRD

BOOLQ (BOOLEAN QUESTION) TASK

Context →	Normal force -- In a simple case such as an object resting upon a table, the normal force on the object is equal but in opposite direction to the gravitational force applied on the object (or the weight of the object), that is, N = m g (\displaystyle N=mg), where m is mass, and g is the gravitational field strength (about 9.81 m/s on Earth). The normal force here represents the force applied by the table against the object that prevents it from sinking through the table and requires that the table is sturdy enough to deliver this normal force without breaking. However, it is easy to assume that the normal force and weight are action-reaction force pairs (a common mistake). In this case, the normal force and weight need to be equal in magnitude to explain why there is no upward acceleration of the object. For example, a ball that bounces upwards accelerates upwards because the normal force acting on the ball is larger in magnitude than the weight of the ball. question: is the normal force equal to the force of gravity? answer:
Target Completion →	yes

WIC (WORD IN CONTEXT) TASK

Context →	An outfitter provided everything needed for the safari. Before his first walking holiday, he went to a specialist outfitter to buy some boots. question: Is the word 'outfitter' used in the same way in the two sentences above? answer:
Target Completion →	no

COPA TASK

Context →	My body cast a shadow over the grass because
Correct Answer →	the sun was rising.
Incorrect Answer →	the grass was cut.

WSC (WINOGRAD SCHEMA CHALLENGE) TASK

Context →	Final Exam with Answer Key Instructions: Please carefully read the following passages. For each passage, you must identify which noun the pronoun marked in *bold* refers to. ===== Passage: Mr. Moncrieff visited Chester's luxurious New York apartment, thinking that it belonged to his son Edward. The result was that Mr. Moncrieff has decided to cancel Edward's allowance on the ground that he no longer requires *his* financial support. Question: In the passage above, what does the pronoun "*his*" refer to? Answer:
Target Completion →	mr. moncrieff

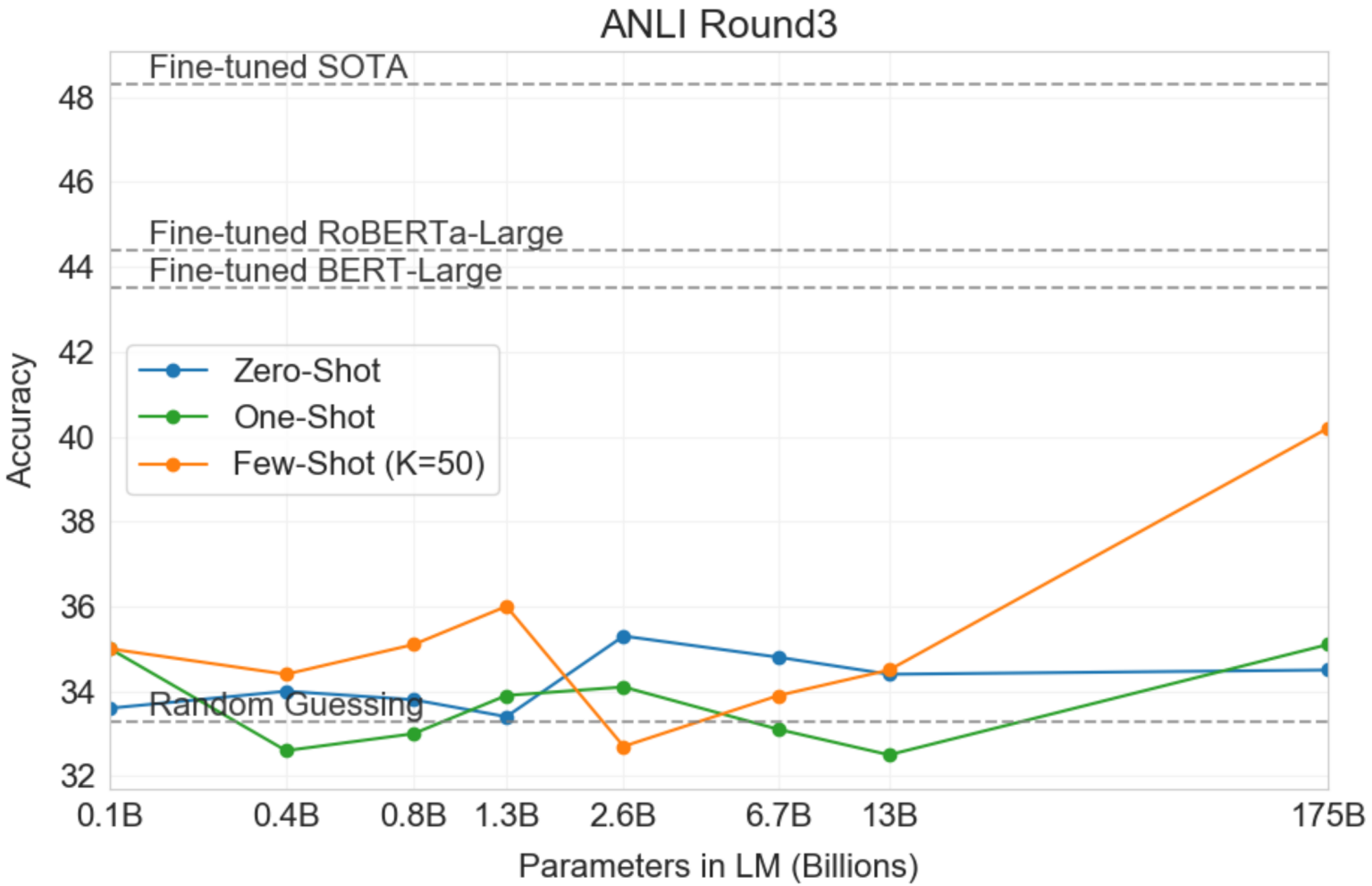
RECORD TASK

Context →	(CNN) Yuval Rabin, whose father, Yitzhak Rabin, was assassinated while serving as Prime Minister of Israel, criticized Donald Trump for appealing to "Second Amendment people" in a speech and warned that the words that politicians use can incite violence and undermine democracy. "Trump's words are an incitement to the type of political violence that touched me personally," Rabin wrote in USAToday. He said that Trump's appeal to "Second Amendment people" to stop Hillary Clinton -- comments that were criticized as a call for violence against Clinton, something Trump denied -- "were a new level of ugliness in an ugly campaign season." - The son of a former Israeli Prime Minister who was assassinated wrote an op ed about the consequence of violent political rhetoric. - Warns of "parallels" between Israel of the 1990s and the U.S. today.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Donald Trump's aggressive rhetoric.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Trump's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Hillary Clinton's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned U.S.'s aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Yitzhak Rabin's aggressive rhetoric.

ANLI TASK

Context →	anli 3: anli 3: We shut the loophole which has American workers actually subsidizing the loss of their own job. They just passed an expansion of that loophole in the last few days: $43 billion of giveaways, including favors to the oil and gas industry and the people importing ceiling fans from China. Question: The loophole is now gone True, False, or Neither?
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

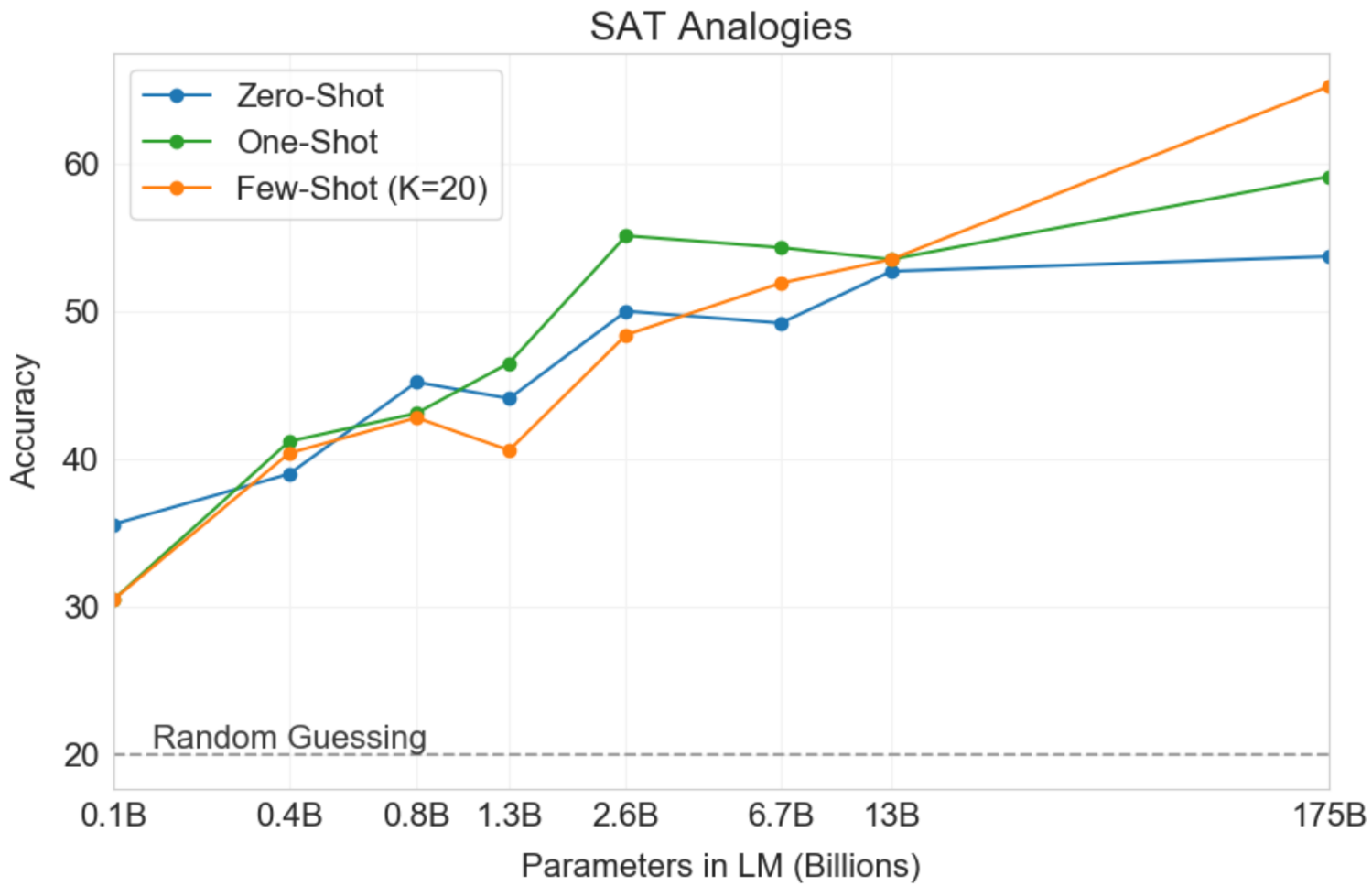
PERFORMANCE ON ANLI TASK



SAT ANALOGIES TASK

Context →	lull is to trust as
Correct Answer →	cajole is to compliance
Incorrect Answer →	balk is to fortitude
Incorrect Answer →	betray is to loyalty
Incorrect Answer →	hinder is to destination
Incorrect Answer →	soothe is to passion

PERFORMANCE ON SAT ANALOGIES



GPT3 CAN CORRECT GRAMMAR

Poor English input: I eated the purple berries.
Good English output: I ate the purple berries.
Poor English input: Thank you for picking me as your designer. I'd appreciate it.
Good English output: Thank you for choosing me as your designer. I appreciate it.
Poor English input: The mentioned changes have done. or I did the alteration that you requested. or I changed things you wanted and did the modifications.
Good English output: The requested changes have been made. or I made the alteration that you requested. or I changed things you wanted and made the modifications.
Poor English input: I'd be more than happy to work with you in another project.
Good English output: I'd be more than happy to work with you on another project.

Poor English input: Please provide me with a short brief of the design you're looking for and that'd be nice if you could share some examples or project you did before.
Good English output: Please provide me with a brief description of the design you're looking for and that would be nice if you could share some examples or projects you have done before.

Poor English input: The patient was died.
Good English output: The patient died.

Poor English input: We think that Leslie likes ourselves.
Good English output: We think that Leslie likes us.

Poor English input: Janet broke Bill on the finger.
Good English output: Janet broke Bill's finger.

Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap.
Good English output: Mary arranged for John to rent a house in St. Louis.

GENERATION OF NEWS ARTICLES

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- Context given to gpt3:

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- Context given to gpt3:
  - Three “training” articles to condition gpt3

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  - Title and subtitle of a 4th article

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- Evaluation: Humans are presented the human-generated original article and the gpt3-generated article and are asked to identify which is fake.

HUMANS CANNOT DISTINGUISH HUMAN GENERATED VS GPT-3 GENERATED

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p-value)	"I don't know" assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 (2e-4)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 (7e-21)	6.0%
GPT-3 Large	68%	64%–72%	7.3 (3e-11)	8.7%
GPT-3 XL	62%	59%–65%	10.7 (1e-19)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 (5e-19)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 (3e-21)	6.2%
GPT-3 13B	55%	52%–58%	15.3 (1e-32)	7.1%
GPT-3 175B	52%	49%–54%	16.9 (1e-34)	7.8%

HARD TO IDENTIFY AS FAKE

Title: United Methodists Agree to Historic Split
Subtitle: Those who oppose gay marriage will form their own denomination
Article: After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.
The Post notes that the denomination, which claims 12.5 million members, was in the early 20th century the "largest Protestant denomination in the U.S.," but that it has been shrinking in recent decades. The new split will be the second in the church's history. The first occurred in 1968, when roughly 10 percent of the denomination left to form the Evangelical United Brethren Church. The Post notes that the proposed split "comes at a critical time for the church, which has been losing members for years," which has been "pushed toward the brink of a schism over the role of LGBTQ people in the church." Gay marriage is not the only issue that has divided the church. In 2016, the denomination was split over ordination of transgender clergy, with the North Pacific regional conference voting to ban them from serving as clergy, and the South Pacific regional conference voting to allow them.

EASIER TO IDENTIFY AS FAKE

Title: Star's Tux Promise Draws Megyn Kelly's Sarcasm
Subtitle: Joaquin Phoenix pledged to not change for each awards event
Article: A year ago, Joaquin Phoenix made headlines when he appeared on the red carpet at the Golden Globes wearing a tuxedo with a paper bag over his head that read, "I am a shape-shifter. I can't change the world. I can only change myself." It was a promise to not change to fit into the Hollywood mold: "I think that's a really special thing, to not change yourself. I think it's a really special thing to say, 'This is what's inside of me, I'm proud of it, and I'm not going to be ashamed because of the way that someone else thinks I should be.'" Now, it's the Oscars, and Phoenix is at it again. But this time, his publicist is saying he'll be wearing a tux no matter what.
Megyn Kelly was not impressed, and she let him have it on The Tonight Show. "You know, I feel like, I feel like you could have worn the tux," she says. "But you're saying you're a shape-shifter. I don't know if you can change your tux, but you can change your mind. You can change your mind. You can change your mind." Phoenix says he did, but it didn't stick. "I was like, 'Okay, I'm going to wear a tuxedo to this thing.' And then I thought, 'I don't want to wear a tuxedo to this thing.'" Kelly goes on to encourage him to change his mind again, but Phoenix says it's too late: "I'm committed to wearing this."

SUMMARY

- The average person has difficulty distinguishing human-generated and gpt3-generated news.
- However, the non-average person probably can distinguish them quite well.
- It is unclear whether machines are able to distinguish human-generated from gpt3-generated news.
- This has great significance for preventing abuse of AI technology.